KAIZEN MATHS

Lesson starter

Midpoint and Length of Line Segments

Midpoint

Name _____

S1

Find the midpoint of A (-1, -4) and B (0, -5) .

S2

M (1, 1) is the midpoint of AB. A is (0, 2) . Find B.

S3

Find the midpoint of A (-1, -4) and B (-1, 1) .

S4

M (-3, -2) is the midpoint of AB. A is (-1, -1) . Find B.

Show your working

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GEO-3384-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Midpoint

Name _____

E1

Find the midpoint of A (2, 5) and B (4, 4) .

E2

M (2, 0) is the midpoint of AB. A is (-1, 5) . Find B.

E3

Find the midpoint of A (-5, -1) and B (-3, 3) .

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Midpoint

Name _____

E1

Find the midpoint of A (2, 5) and B (4, 4) .

E2

M (2, 0) is the midpoint of AB. A is (-1, 5) . Find B.

E3

Find the midpoint of A (-5, -1) and B (-3, 3) .

GEO-3384-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Midpoint

Name _____

E1

Find the midpoint of A (2, 5) and B (4, 4) .

E2

M (2, 0) is the midpoint of AB. A is (-1, 5) . Find B.

E3

Find the midpoint of A (-5, -1) and B (-3, 3) .

GEO-3384-L1 · E

Show your working

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Midpoint and Length of Line Segments

Midpoint

Name _____

E1

Find the midpoint of A (2, 5) and B (4, 4) .

E2

M (2, 0) is the midpoint of AB. A is (-1, 5) . Find B.

E3

Find the midpoint of A (-5, -1) and B (-3, 3) .

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Show your working

Teacher answers & working

Midpoint and Length of Line Segments · Midpoint

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S1

Find the midpoint of A (-1, -4) and B (0, -5) .

Answer:

$(\frac{-1}{2}, \frac{-9}{2})$

Working:

Use the midpoint formula. Average the x-coordinates and average the y-coordinates separately.

$M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$

$M = (\frac{-1+0}{2}, \frac{-4+-5}{2})$

$M = (\frac{-1}{2}, \frac{-9}{2})$

The midpoint is $(\frac{-1}{2}, \frac{-9}{2})$.

S2

M (1, 1) is the midpoint of AB. A is (0, 2) . Find B.

Answer:

$(2, 0)$

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(1) - (0) = 2$

$y_B = 2y_M - y_A = 2(1) - (2) = 0$

Therefore B = (2, 0).

S3

Find the midpoint of A (-1, -4) and B (-1, 1) .

Answer:

$(-1, \frac{-3}{2})$

Working:

Use the midpoint formula. Average the x-coordinates and average the y-coordinates separately.

$M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$

$M = (\frac{-1+-1}{2}, \frac{-4+1}{2})$

$M = (-1, \frac{-3}{2})$

The midpoint is $(-1, \frac{-3}{2})$.

Teacher answers & working

Midpoint and Length of Line Segments · Midpoint

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S4

M (-3, -2) is the midpoint of AB. A is (-1, -1) . Find B.

Answer:

(-5, -3)

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(-3) - (-1) = -5$

$y_B = 2y_M - y_A = 2(-2) - (-1) = -3$

Therefore B = (-5, -3).

E1

Find the midpoint of A (2, 5) and B (4, 4) .

Answer:

$(3, \frac{9}{2})$

Working:

Use the midpoint formula. Average the x-coordinates and average the y-coordinates separately.

$M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$

$M = (\frac{2+4}{2}, \frac{5+4}{2})$

$M = (3, \frac{9}{2})$

The midpoint is $(3, \frac{9}{2})$.

E2

M (2, 0) is the midpoint of AB. A is (-1, 5) . Find B.

Answer:

(5, -5)

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(2) - (-1) = 5$

$y_B = 2y_M - y_A = 2(0) - (5) = -5$

Therefore B = (5, -5).

Teacher answers & working

Midpoint and Length of Line Segments · Midpoint

GEO-3384-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Find the midpoint of A (-5, -1) and B (-3, 3) .

Answer:

(-4, 1)

Working:

Use the midpoint formula. Average the x-coordinates and average the y-coordinates separately.

$M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$

$M = (\frac{-5+-3}{2}, \frac{-1+3}{2})$

$M = (-4, 1)$

The midpoint is (-4, 1).